

COURSE CODE/TITLE:	CPE212 - Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	07/16/2026	DATE SUBMITTED:	07/16/2026
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ACTIVITY NO. 1.1 Creating Virtual Machines

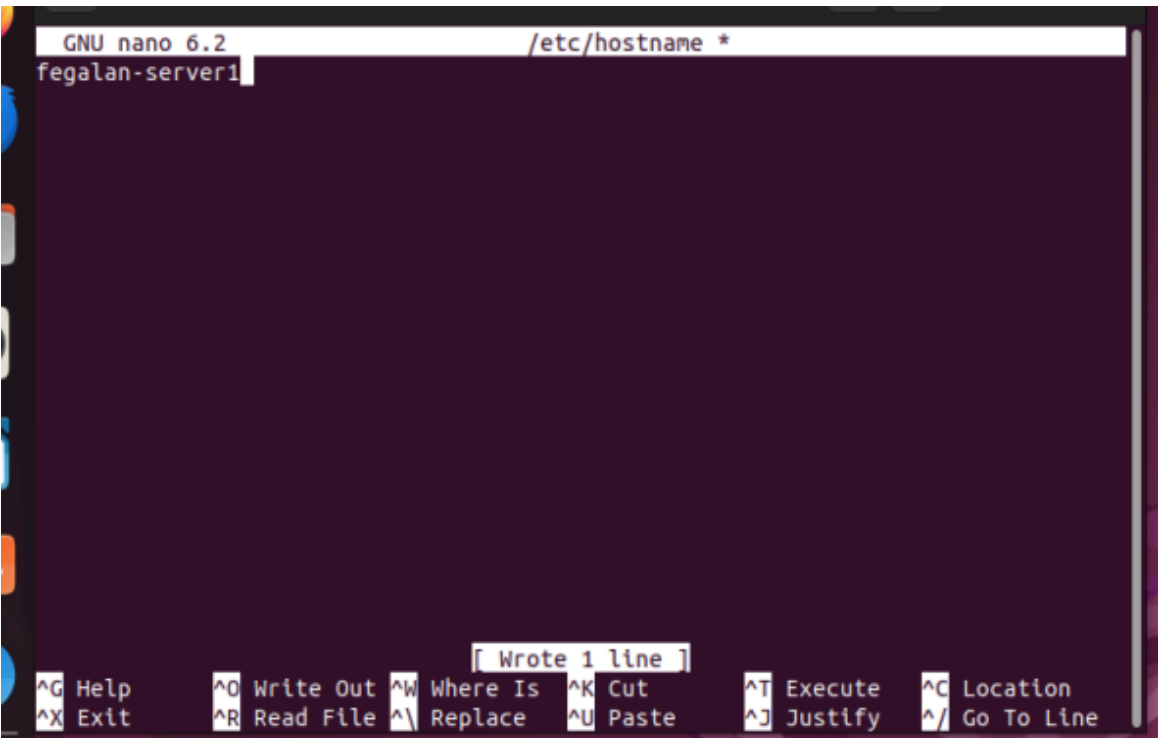
1. PROCEDURE OUTPUT

Task 1: Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press control + O to write out (save the file). Press enter when asked for the name of the file. Press control + X to end.

1. Change the hostname using the command `sudo nano /etc/hostname`

1.1 Use server1 for Server 1

```
fegalanvm@fegalanvm-VirtualBox:~$ sudo nano /etc/hostname
[sudo] password for fegalanvm:
fegalanvm@fegalanvm-VirtualBox:~$ sudo nano /etc/hosts
fegalanvm@fegalanvm-VirtualBox:~$
```



- Changed the hostname using the commands as instructed.

1.2 Use server2 for Server 2

```
fegalanvm@fegalanvm-VirtualBox: ~
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

fegalanvm@fegalanvm-VirtualBox:~$ sudo nano /etc/hostname
[sudo] password for fegalanvm:
fegalanvm@fegalanvm-VirtualBox:~$ sudo nano /etc/hosts
fegalanvm@fegalanvm-VirtualBox:~$ cat /etc/hosts
127.0.0.1    localhost
127.0.1.1    fegalan-server2

# The following lines are desirable for IPv6 capable hosts
::1        ip6-localhost ip6-loopback
fe00::0    ip6-localnet
ff00::0    ip6-mcastprefix
ff02::1    ip6-allnodes
ff02::2    ip6-allrouters
fegalanvm@fegalanvm-VirtualBox:~$ cat /etc/hostname
fegalanvm-server2
fegalanvm@fegalanvm-VirtualBox:~$ s
```

- Changed the hostname using the commands as instructed.

1.3 Use workstation for the Local Machine

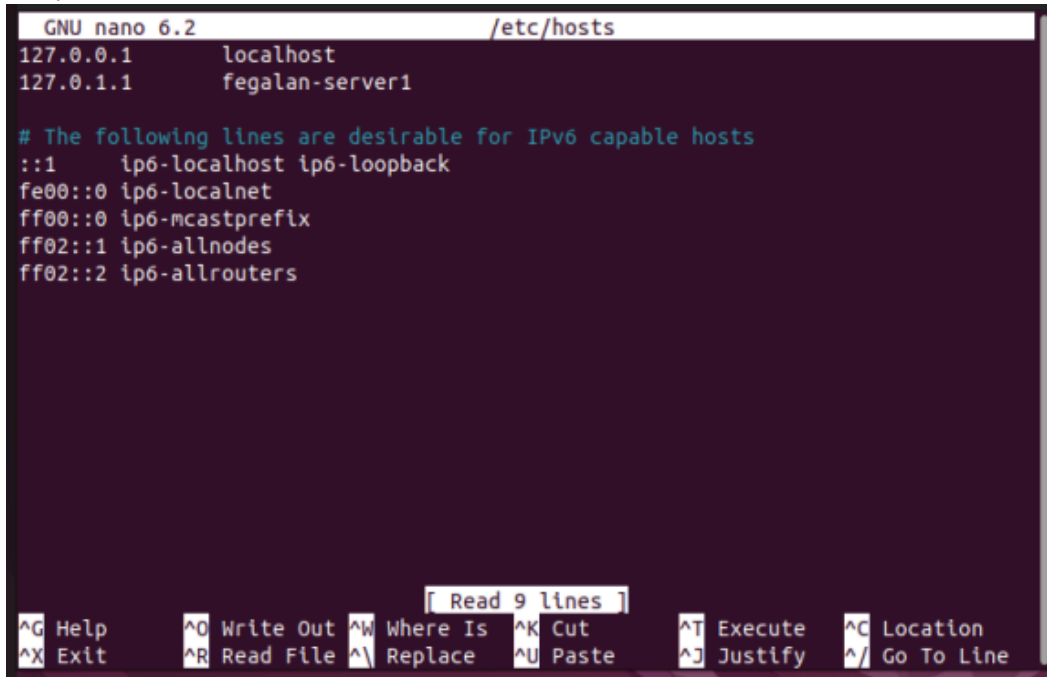
```
fegalanvm@fegalan-workstation: ~
fegalanvm@fegalan-workstation:~$ cat /etc/hostname
fegalan-workstation
fegalanvm@fegalan-workstation:~$ cat /etc/hosts
127.0.0.1    localhost
127.0.1.1    fegalan-workstation

# The following lines are desirable for IPv6 capable hosts
::1        ip6-localhost ip6-loopback
fe00::0    ip6-localnet
ff00::0    ip6-mcastprefix
ff02::1    ip6-allnodes
ff02::2    ip6-allrouters
fegalanvm@fegalan-workstation:~$
```

- Changed the hostname using the commands as instructed.

2. Edit the hosts using the command `sudo nano /etc/hosts`. Edit the second line.

2.1 Type 127.0.0.1 server 1 for Server 1



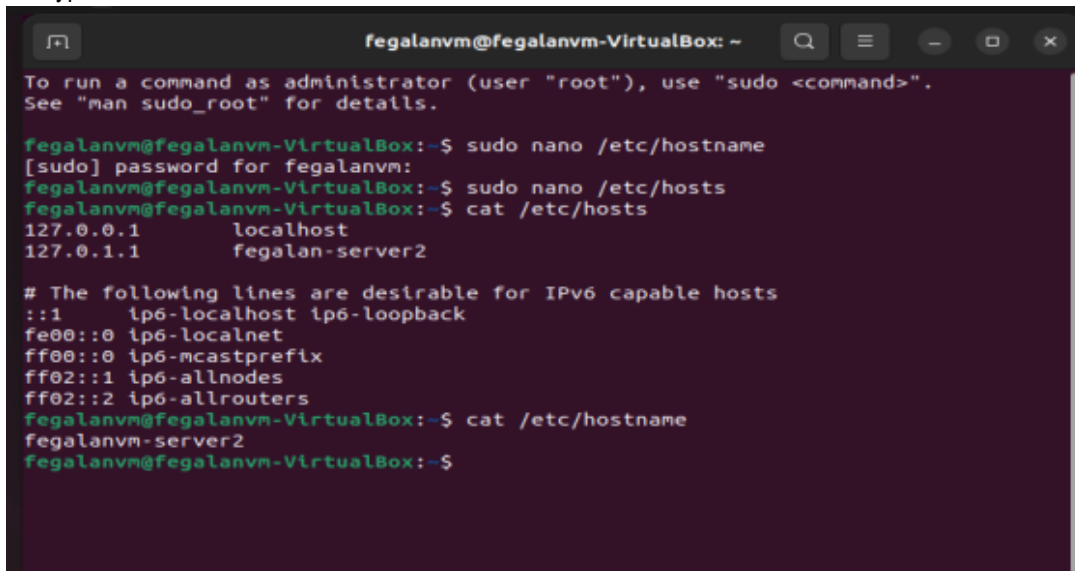
```
GNU nano 6.2 /etc/hosts
127.0.0.1    localhost
127.0.1.1    fegalan-server1

# The following lines are desirable for IPv6 capable hosts
::1         ip6-localhost ip6-loopback
fe00::0     ip6-localnet
ff00::0     ip6-mcastprefix
ff02::1     ip6-allnodes
ff02::2     ip6-allrouters

Read 9 lines
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^_ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

- Opened the `/etc/hosts` and edited the second line from the old name to `fegalan-server1`

2.2 Type 127.0.0.1 server 2 for Server 2



```
fegalanvm@fegalanvm-VirtualBox: ~
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

fegalanvm@fegalanvm-VirtualBox:~$ sudo nano /etc/hostname
[sudo] password for fegalanvm:
fegalanvm@fegalanvm-VirtualBox:~$ sudo nano /etc/hosts
fegalanvm@fegalanvm-VirtualBox:~$ cat /etc/hosts
127.0.0.1    localhost
127.0.1.1    fegalan-server2

# The following lines are desirable for IPv6 capable hosts
::1         ip6-localhost ip6-loopback
fe00::0     ip6-localnet
ff00::0     ip6-mcastprefix
ff02::1     ip6-allnodes
ff02::2     ip6-allrouters
fegalanvm@fegalanvm-VirtualBox:~$ cat /etc/hostname
fegalanvm-server2
fegalanvm@fegalanvm-VirtualBox:~$
```

- Opened the `/etc/hosts` and edited the second line from the old name to `fegalan-server2`

2.3 Type 127.0.0.1 workstation for the Local Machine

```

fes
fegalanvm@fegalan-workstation: ~
fegalanvm@fegalan-workstation:~$ cat /etc/hostname
fegalan-workstation
fegalanvm@fegalan-workstation:~$ cat /etc/hosts
127.0.0.1    localhost
127.0.1.1    fegalan-workstation

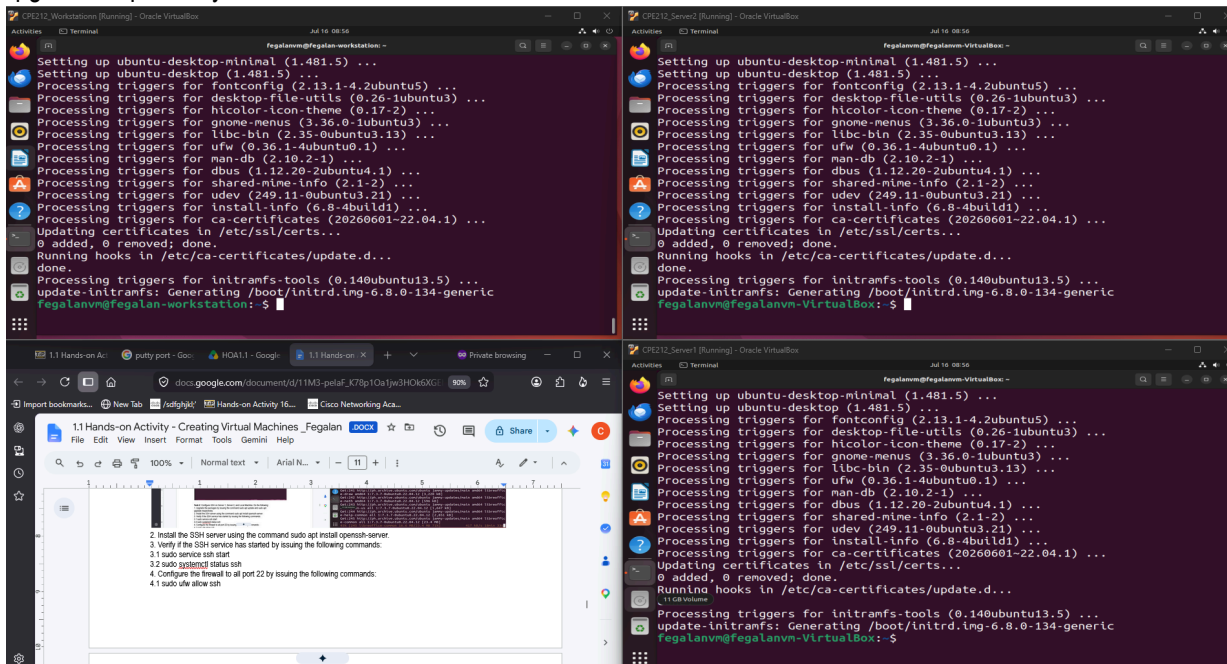
# The following lines are desirable for IPv6 capable hosts
::1        ip6-localhost ip6-loopback
fe00::0    ip6-localnet
ff00::0    ip6-mcastprefix
ff02::1    ip6-allnodes
ff02::2    ip6-allrouters
fegalanvm@fegalan-workstation:~$

```

- Opened the /etc/hosts and edited the second line from the old name to fegalan-workstation

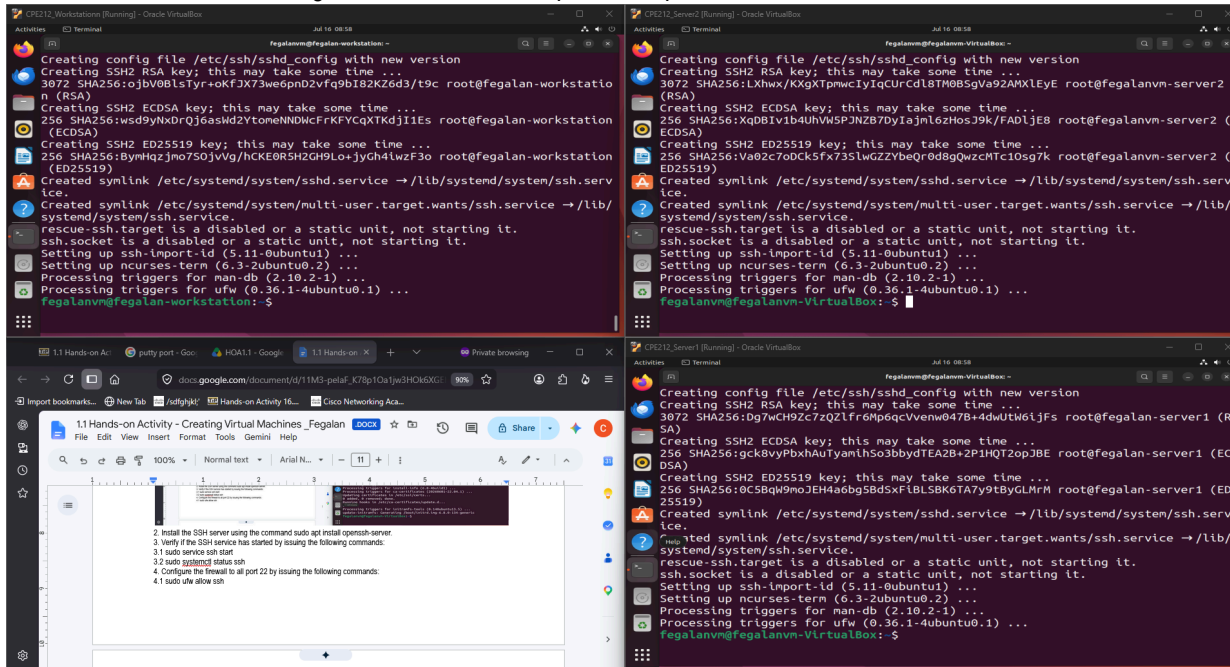
Task 2: Configure SSH on Server 1, Server 2, and Local Machine. Do the following:

1. Upgrade the packages by issuing the command `sudo apt update` and `sudo apt upgrade` respectively.



- I did `sudo apt update` && `apt upgrade -y`. Here `-y` is used just to auto yes when prompted and upgrade should run directly after update not prompting me to type it again.

2. Install the SSH server using the command `sudo apt install openssh-server`.



```

Creating config file /etc/ssh/sshd_config with new version
Creating SSH2 RSA key; this may take some time ...
3072 SHA256:ojBV0B1sYtr+oKfJX73we6pn0Zvfq9B182KZ6d3/t9c root@fegalan-workstation (RSA)
256 SHA256:wsd9yXDrQj6aswd2YtomeNNDwCfRKYFCqXTkdJ1IEs root@fegalan-workstation (ECDSA)
Creating SSH2 ECDSA key; this may take some time ...
256 SHA256:BymHqzjJno7SOjVvg/hcKE0RSH2GH9Lo+jyG4lwzF3o root@fegalan-workstation (ED25519)
Created symlink /etc/systemd/system/ssh.service → /lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /lib/systemd/system/ssh.service.
rescue-ssh.target is a disabled or a static unit, not starting it.
ssh.socket is a disabled or a static unit, not starting it.
Setting up ssh-import-id (5.11-0ubuntu1) ...
Setting up ncurses-term (6.3-2ubuntu0.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...
fegalanv@fegalan-workstation:~$

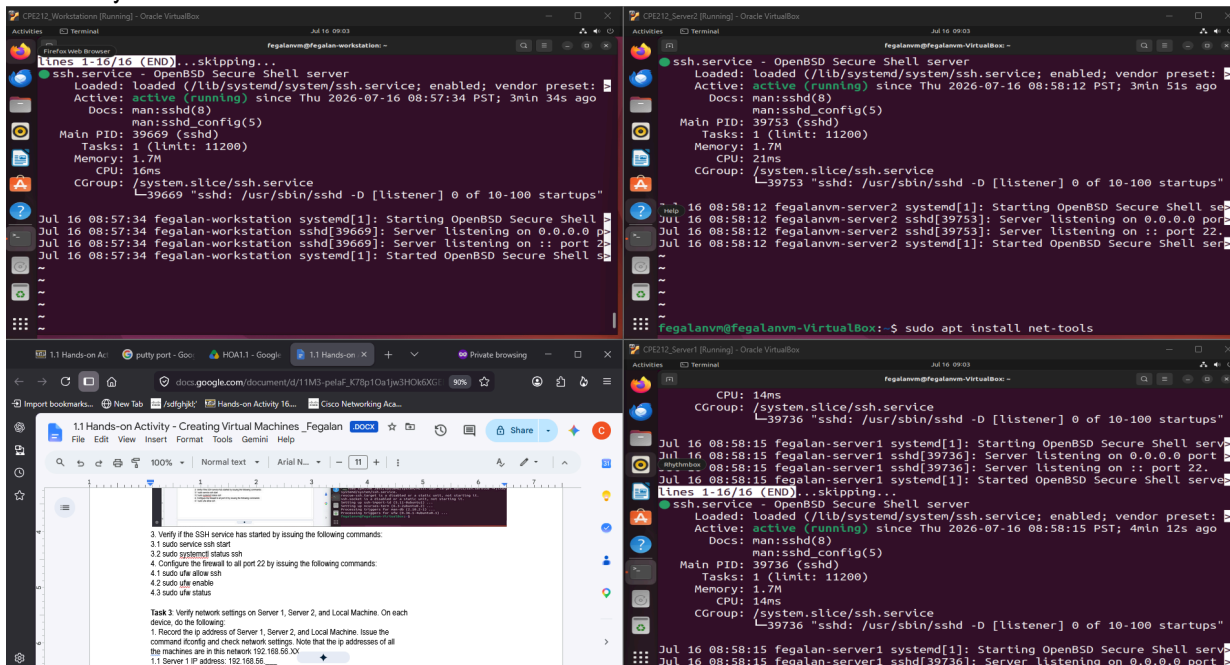
Creating config file /etc/ssh/sshd_config with new version
Creating SSH2 RSA key; this may take some time ...
3072 SHA256:Dg7wCH9ZC7zQZLfr6MpqcvVenv0478+4dwUW6jFs root@fegalanv-server2 (RSA)
256 SHA256:Xq0B1v1b4UHVMPJNZB70yIajnl6zHosJ9K/FADljE8 root@fegalanv-server2 (ECDSA)
Creating SSH2 ECDSA key; this may take some time ...
256 SHA256:Va02c7oDck5fx73SLwGZZYbeQr0d8GqWzCH1C10sg7k root@fegalanv-server2 (ED25519)
Created symlink /etc/systemd/system/ssh.service → /lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /lib/systemd/system/ssh.service.
rescue-ssh.target is a disabled or a static unit, not starting it.
ssh.socket is a disabled or a static unit, not starting it.
Setting up ssh-import-id (5.11-0ubuntu1) ...
Setting up ncurses-term (6.3-2ubuntu0.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...
fegalanv@fegalanv-VirtualBox:~$
  
```

- Installed openssh-server on all vms.

3. Verify if the SSH service has started by issuing the following commands:

3.1 `sudo service ssh start`

3.2 `sudo systemctl status ssh`



```

ssh.service - OpenBSD Secure Shell server
Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
Active: active (running) since Thu 2026-07-16 08:57:34 PST; 3min 34s ago
Docs: man:sshd(8)
      man:sshd_config(5)
Main PID: 39669 (sshd)
Tasks: 1 (limit: 11200)
Memory: 1.7M
CPU: 16ms
CGroup: /system.slice/ssh.service
        └─39669 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 16 08:57:34 fegalan-workstation systemd[1]: Starting OpenBSD Secure Shell server: sshd[39669]: Server listening on 0.0.0.0 port 22.
Jul 16 08:57:34 fegalan-workstation sshd[39669]: Server listening on :: port 22.
Jul 16 08:57:34 fegalan-workstation systemd[1]: Started OpenBSD Secure Shell server: sshd[39669].

ssh.service - OpenBSD Secure Shell server
Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
Active: active (running) since Thu 2026-07-16 08:58:12 PST; 4min 51s ago
Docs: man:sshd(8)
      man:sshd_config(5)
Main PID: 39753 (sshd)
Tasks: 1 (limit: 11200)
Memory: 1.7M
CPU: 21ms
CGroup: /system.slice/ssh.service
        └─39753 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 16 08:58:12 fegalanv-server2 systemd[1]: Starting OpenBSD Secure Shell server: sshd[39753]: Server listening on 0.0.0.0 port 22.
Jul 16 08:58:12 fegalanv-server2 sshd[39753]: Server listening on :: port 22.
Jul 16 08:58:12 fegalanv-server2 systemd[1]: Started OpenBSD Secure Shell server: sshd[39753].

fegalanv@fegalanv-VirtualBox:~$ sudo apt install net-tools

CPU: 14ms
CGroup: /system.slice/ssh.service
        └─39736 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 16 08:58:15 fegalanv-server1 systemd[1]: Starting OpenBSD Secure Shell server: sshd[39736]: Server listening on 0.0.0.0 port 22.
Jul 16 08:58:15 fegalanv-server1 sshd[39736]: Server listening on :: port 22.
Jul 16 08:58:15 fegalanv-server1 systemd[1]: Started OpenBSD Secure Shell server: sshd[39736].

ssh.service - OpenBSD Secure Shell server
Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
Active: active (running) since Thu 2026-07-16 08:58:15 PST; 4min 12s ago
Docs: man:sshd(8)
      man:sshd_config(5)
Main PID: 39736 (sshd)
Tasks: 1 (limit: 11200)
Memory: 1.7M
CPU: 14ms
CGroup: /system.slice/ssh.service
        └─39736 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 16 08:58:15 fegalanv-server1 systemd[1]: Starting OpenBSD Secure Shell server: sshd[39736]: Server listening on 0.0.0.0 port 22.
Jul 16 08:58:15 fegalanv-server1 sshd[39736]: Server listening on :: port 22.
Jul 16 08:58:15 fegalanv-server1 systemd[1]: Started OpenBSD Secure Shell server: sshd[39736].
  
```

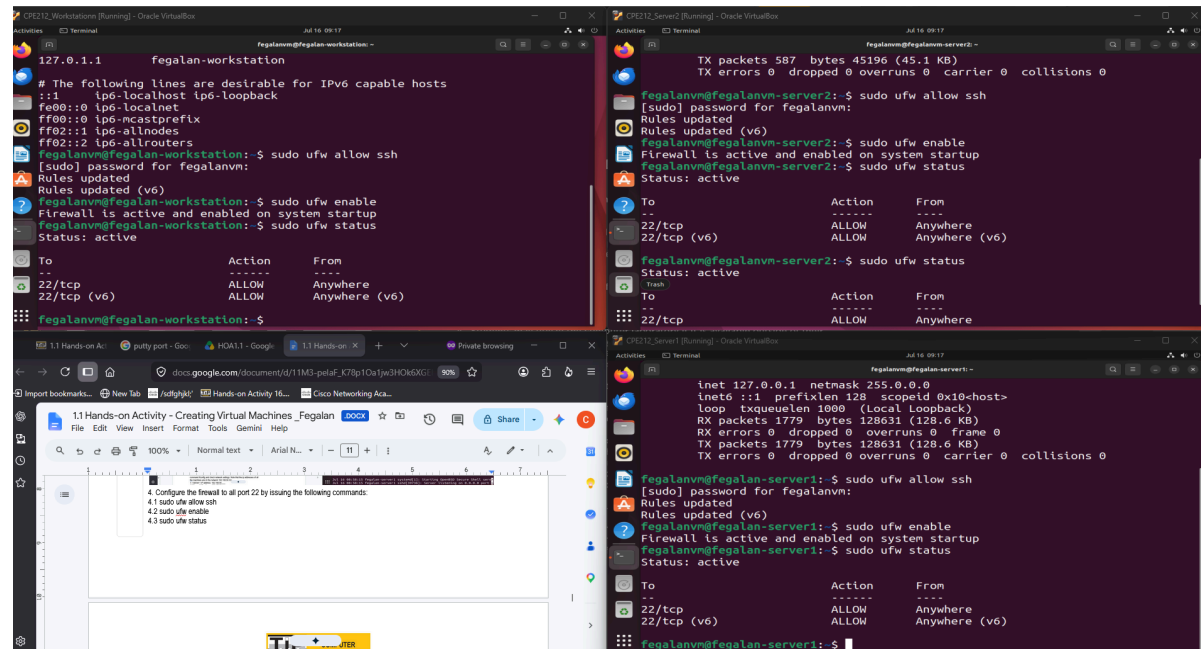
- Here I verified the ssh service as instructed.

4. Configure the firewall to all port 22 by issuing the following commands:

4.1 `sudo ufw allow ssh`

4.2 sudo ufw enable

4.3 sudo ufw status



The screenshot shows a terminal window on a host named 'Fegalan-workstation' with IP 127.0.0.1. The user is configuring the UFW firewall. The steps shown are:

```

# The following lines are desirable for IPv6 capable hosts
:::1 ip6:localhost ip6-loopback
fe80:: ip6-mcastprefix
ff02:: ip6-allnodes
ff02:: ip6-allrouters

fegalanvm@fegalan-workstation:~$ sudo ufw allow ssh
[sudo] password for fegalanvm:
Rules updated
Rules updated (v6)
fegalanvm@fegalan-workstation:~$ sudo ufw enable
Firewall is active and enabled on system startup
fegalanvm@fegalan-workstation:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)

fegalanvm@fegalan-workstation:~$
  
```

Below the terminal, a browser window shows a document titled '11 Hands-on Activity - Creating Virtual Machines_Fegalan'. It contains instructions for configuring the firewall:

```

1. Configure the firewall to allow port 22 by issuing the following commands:
4.1 sudo ufw allow ssh
4.2 sudo ufw enable
4.3 sudo ufw status
  
```

Another terminal window shows the status of the firewall on 'fegalanvm@fegalan-server2':

```

fegalanvm@fegalan-server2:~$ sudo ufw allow ssh
[sudo] password for fegalanvm:
Rules updated
Rules updated (v6)
fegalanvm@fegalan-server2:~$ sudo ufw enable
Firewall is active and enabled on system startup
fegalanvm@fegalan-server2:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)

fegalanvm@fegalan-server2:~$
  
```

- Configured the port as instructed using the ubuntu firewall.

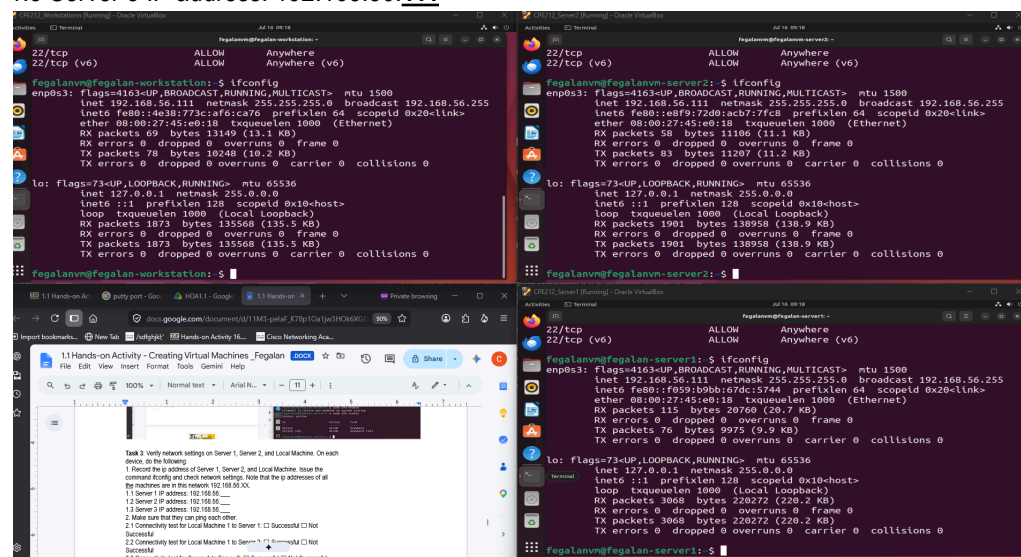
Task 3: Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:

1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command `ifconfig` and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.

1.1 Server 1 IP address: 192.168.56.111

1.2 Server 2 IP address: 192.168.56.111

1.3 Server 3 IP address: 192.168.56.111



The screenshot shows terminal windows on 'fegalanvm@fegalan-workstation' and 'fegalanvm@fegalan-server2' running the `ifconfig` command to verify network settings.

On 'fegalanvm@fegalan-workstation':

```

fegalanvm@fegalan-workstation:~$ ifconfig
enp83: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.111 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::e4b3:77c5:afca:76 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:45:e0:18 txqueuelen 1000 (Ethernet)
    RX packets 69 bytes 13149 (13.1 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 78 bytes 10248 (10.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1873 bytes 135568 (135.5 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1873 bytes 135568 (135.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

fegalanvm@fegalan-workstation:~$
  
```

On 'fegalanvm@fegalan-server2':

```

fegalanvm@fegalan-server2:~$ ifconfig
enp83: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.111 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::e4b3:77c5:afca:76 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:45:e0:18 txqueuelen 1000 (Ethernet)
    RX packets 58 bytes 11106 (11.1 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 83 bytes 11207 (11.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1901 bytes 138958 (138.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1901 bytes 138958 (138.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

fegalanvm@fegalan-server2:~$
  
```

Below the terminal windows, a browser window shows a document titled '11 Hands-on Activity - Creating Virtual Machines_Fegalan'. It contains instructions for Task 3:

```

Task 3: Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:
1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command ifconfig and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.
1.1 Server 1 IP address: 192.168.56.111
1.2 Server 2 IP address: 192.168.56.111
1.3 Server 3 IP address: 192.168.56.111
2. Make sure that they can ping each other.
2.1 Connectivity test for Local Machine to Server 1: [S] Successful [X] Not Successful
2.2 Connectivity test for Local Machine to Server 2: [S] Successful [X] Not Successful
2.3 Connectivity test for Server 1 to Server 2: [S] Successful [X] Not Successful
  
```

- Just put this screenshot as proof of the ips of each vm.

2. Make sure that they can ping each other.

2.1 Connectivity test for Local Machine 1 to Server 1: **Successful**

```
fegalanvm@fegalan-workstation:~$ ping 192.168.56.112
PING 192.168.56.112 (192.168.56.112) 56(84) bytes of data.
64 bytes from 192.168.56.112: icmp_seq=1 ttl=64 time=1.17 ms
64 bytes from 192.168.56.112: icmp_seq=2 ttl=64 time=3.11 ms
64 bytes from 192.168.56.112: icmp_seq=3 ttl=64 time=2.17 ms
64 bytes from 192.168.56.112: icmp_seq=4 ttl=64 time=1.51 ms
64 bytes from 192.168.56.112: icmp_seq=5 ttl=64 time=1.77 ms
64 bytes from 192.168.56.112: icmp_seq=6 ttl=64 time=1.44 ms
64 bytes from 192.168.56.112: icmp_seq=7 ttl=64 time=1.25 ms
64 bytes from 192.168.56.112: icmp_seq=8 ttl=64 time=2.24 ms
64 bytes from 192.168.56.112: icmp_seq=9 ttl=64 time=1.17 ms
```

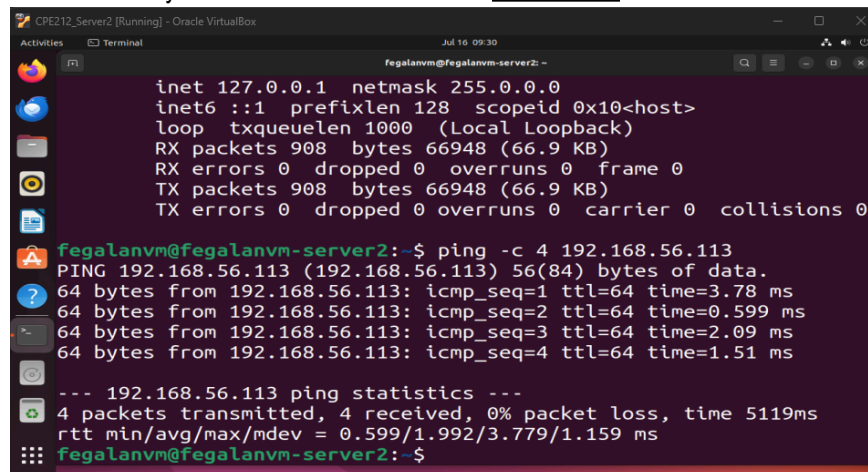
- Pinging successful

2.2 Connectivity test for Local Machine 1 to Server 2: **Successful**

```
fegalanvm@fegalan-workstation:~$ ping 192.168.56.113
PING 192.168.56.113 (192.168.56.113) 56(84) bytes of data.
64 bytes from 192.168.56.113: icmp_seq=1 ttl=64 time=0.867 ms
64 bytes from 192.168.56.113: icmp_seq=2 ttl=64 time=2.19 ms
64 bytes from 192.168.56.113: icmp_seq=3 ttl=64 time=4.32 ms
64 bytes from 192.168.56.113: icmp_seq=4 ttl=64 time=1.87 ms
^C
--- 192.168.56.113 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 7411ms
rtt min/avg/max/mdev = 0.867/2.309/4.316/1.256 ms
```

- Pinging successful

2.3 Connectivity test for Server 1 to Server 2: **Successful**



```
CPE212_Server2 [Running] - Oracle VirtualBox
Activities Terminal Jul 16 09:30
fegalanvm@fegalanvm-server2:~$ ifconfig
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 908 bytes 66948 (66.9 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 908 bytes 66948 (66.9 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

fegalanvm@fegalanvm-server2:~$ ping -c 4 192.168.56.113
PING 192.168.56.113 (192.168.56.113) 56(84) bytes of data.
64 bytes from 192.168.56.113: icmp_seq=1 ttl=64 time=3.78 ms
64 bytes from 192.168.56.113: icmp_seq=2 ttl=64 time=0.599 ms
64 bytes from 192.168.56.113: icmp_seq=3 ttl=64 time=2.09 ms
64 bytes from 192.168.56.113: icmp_seq=4 ttl=64 time=1.51 ms

--- 192.168.56.113 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 5119ms
rtt min/avg/max/mdev = 0.599/1.992/3.779/1.159 ms
fegalanvm@fegalanvm-server2:~$
```

- Pinging successful

Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.

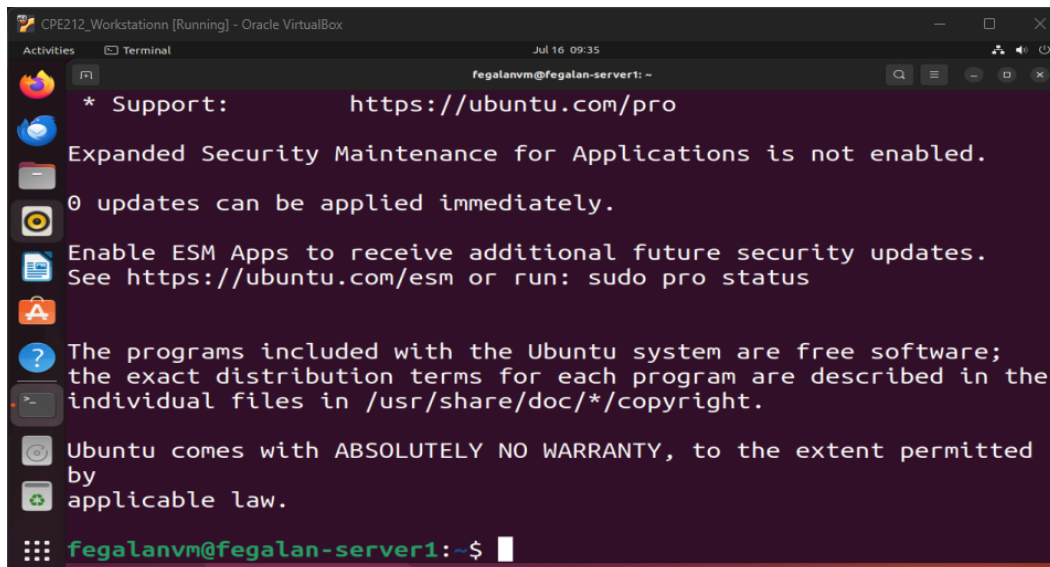
1. On the Local Machine, issue the following commands:

1.1 ssh username@ip_address_server1 for example, ssh jvtaylor@192.168.56.120

1.2 Enter the password for server 1 when prompted

1.3 Verify that you are in server 1. The user should be in this format user@server1.

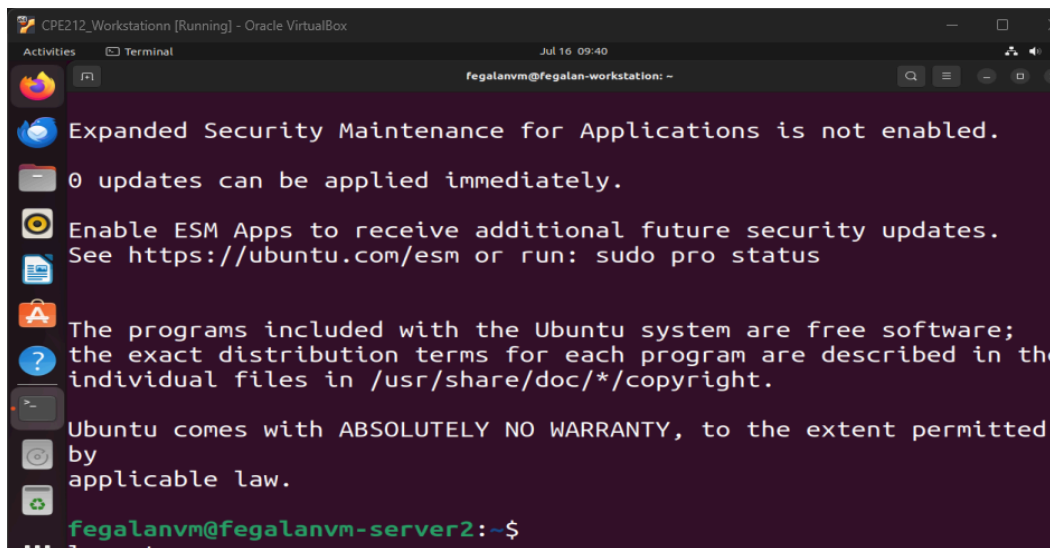
For example, jvtaylor@server1



The screenshot shows a terminal window titled "CPE212_Workstationn [Running] - Oracle VirtualBox". The terminal displays the login prompt for "fegalanvm@fegalan-server1: ~". The output of the login process includes a support link, a message about Expanded Security Maintenance (ESM), update information, ESM instructions, software license information, and a warranty disclaimer. The prompt ends with "fegalanvm@fegalan-server1:~\$".

```
* Support:      https://ubuntu.com/pro
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted
by applicable law.
fegalanvm@fegalan-server1:~$
```

- Verified the ssh connectivity as instructed followed it step by step. This shows the workstation being connected to server 1.
2. Logout of Server 1 by issuing the command control + D.
 3. Do the same for Server 2.



The screenshot shows a terminal window titled "CPE212_Workstationn [Running] - Oracle VirtualBox". The terminal displays the login prompt for "fegalanvm@fegalan-workstation: ~". The output of the login process is identical to the first screenshot, showing the support link, ESM message, update information, ESM instructions, software license information, and warranty disclaimer. The prompt ends with "fegalanvm@fegalanvm-server2:~\$".

```
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted
by applicable law.
fegalanvm@fegalanvm-server2:~$
```

- Verified the ssh connectivity as instructed followed it step by step. This shows the workstation being connected to server 2.
4. Edit the hosts of the Local Machine by issuing the command `sudo nano /etc/hosts`. Below all texts type the following:
 - 4.1 IP_address server 1 (provide the ip address of server 1 followed by the hostname)
 - 4.2 IP_address server 2 (provide the ip address of server 2 followed by the hostname)
 - 4.3 Save the file and exit.

```

Connection to 192.168.56.112 closed.
fegalanvm@fegalan-workstation:~$ sudo nano /etc/hosts
[sudo] password for fegalanvm:
fegalanvm@fegalan-workstation:~$ cat /etc/hosts
127.0.0.1        localhost
127.0.1.1        fegalan-workstation
192.168.56.113   fegalan-server1
192.168.56.112   fegalan-server2

# The following lines are desirable for IPv6 capable hosts
::1             ip6-localhost ip6-loopback
fe00::0         ip6-localnet
ff00::0         ip6-mcastprefix
ff02::1         ip6-allnodes
ff02::2         ip6-allrouters

```

- Put the hostname and ip of server1 and server 2 in the vm of workstation /etc/hosts.

5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do `ssh jvtaylor@server1`. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2

```

fegalanvm@fegalan-workstation:~$ ssh fegalanvm@fegalan-server1
The authenticity of host 'fegalan-server1 (192.168.56.113)' can't be established.
ED25519 key fingerprint is SHA256:0C5BqW9moJEH4a6bg5Bd5xF1BLSBK6TA7y9tByGLMrM.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'fegalan-server1' (ED25519) to the list of known hosts.
fegalanvm@fegalan-server1's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

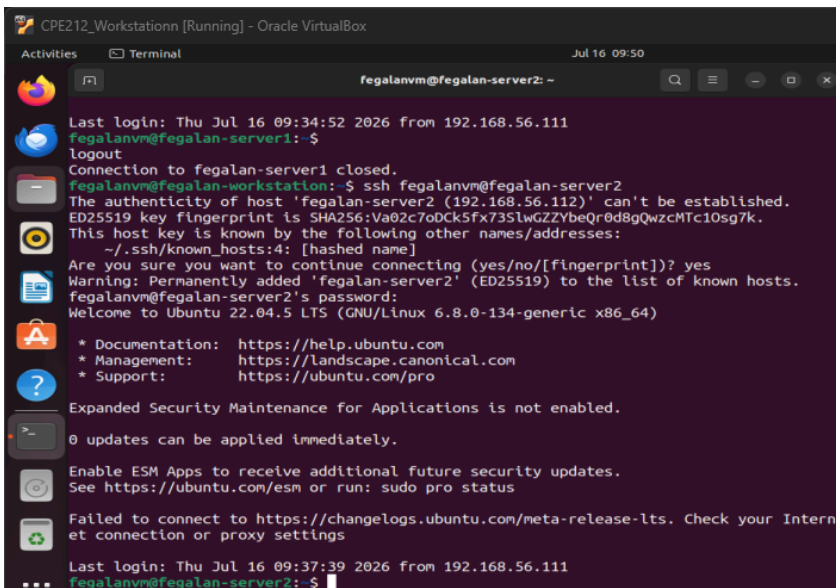
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Thu Jul 16 09:34:52 2026 from 192.168.56.111

```



```

CPE212_Workstationn [Running] - Oracle VirtualBox
Activities Terminal Jul 16 09:50
fegalanvm@fegalan-server2: ~
Last login: Thu Jul 16 09:34:52 2026 from 192.168.56.111
fegalanvm@fegalan-server1:~$
logout
Connection to fegalan-server1 closed.
fegalanvm@fegalan-workstation:~$ ssh fegalanvm@fegalan-server2
The authenticity of host 'fegalan-server2 (192.168.56.112)' can't be established.
ED25519 key fingerprint is SHA256:Va02c7oDCK5fx73SlwGZZYbeQr0d8gQwzcMTC10sg7k.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'fegalan-server2' (ED25519) to the list of known hosts.
fegalanvm@fegalan-server2's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Thu Jul 16 09:37:39 2026 from 192.168.56.111
fegalanvm@fegalan-server2:~$

```

- Checking if the configured /etc/hosts file contains the saved remote_hostname and remote_ip by connecting only using the remote_hostname.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections: (Answer the following)

1. How are we able to use the hostname instead of IP address in SSH commands?

- This is because in etc/hosts we configured it and set the ip of the server vms so it remembers what hostname is tied to that ip when called.

2. How secured is SSH?

- SSH uses a bunch of ways to make connections secure such as having a private key only on your device and a public key that the server has, making the systems communicate with each other. Once connection is established there is a process of decryption where both server and your personal device has a shared secret key just for that specific session and everything you do on your personal device is encrypted and decrypted at the server making it unable to read the traffic through the internet.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Gemini

Purpose of Use:

For grammar correction, and assistance with understanding concepts.

Extent of Use:

For grammar correction i mainly used it to check if how i typed it was correct, suggestions on how i could make to my text to make it better and for understanding of concepts i mainly used it to check for other test cases using that same command such as `remote_username@remote_ip` confirming if whether you use the `remote_username` or the `local username`. Also good practices on how to configure files like `/etc/hosts` how should i format and structure for it.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.